Stacked DRAM

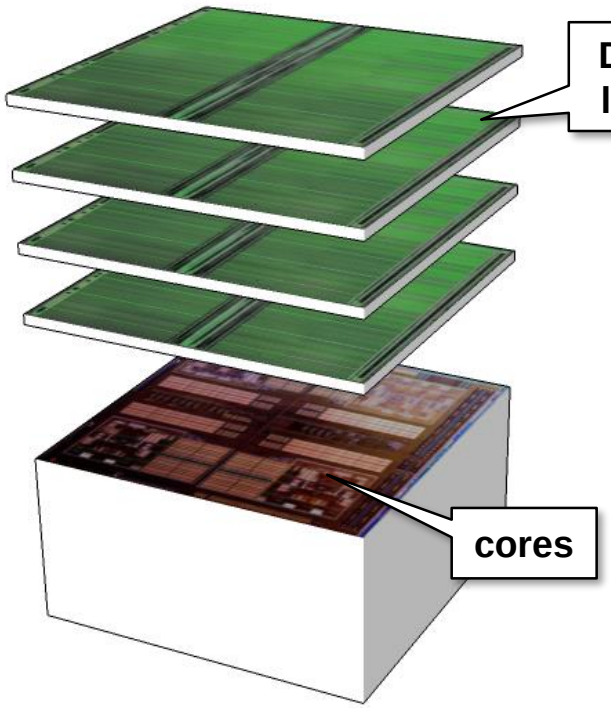
Sparsh Mittal

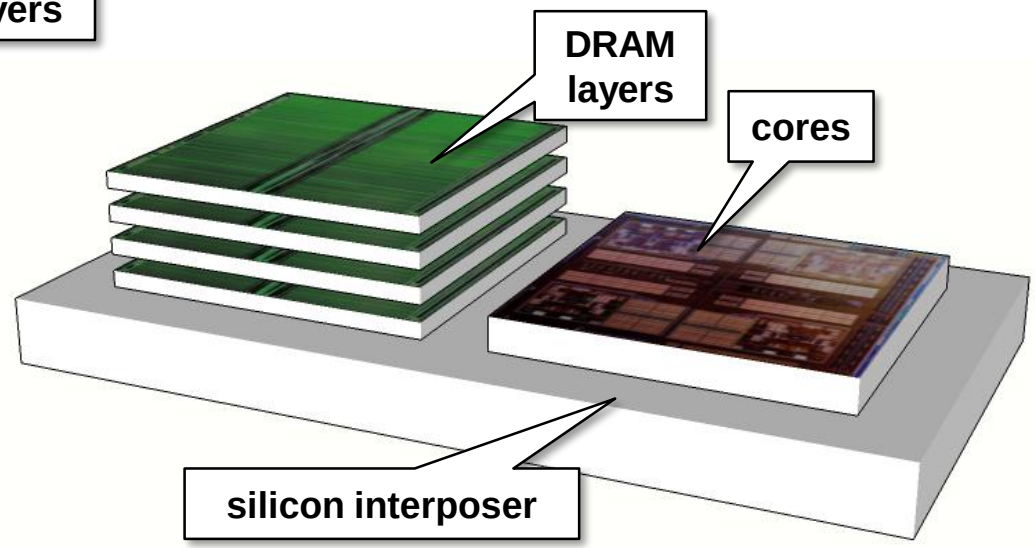
Limitations of SRAM cache

- Low density, high leakage power
- Large caches designed with SRAM can take 90% of chip area in future
- Large leakage power of SRAM: expensive cooling solutions required, e.g., liquid cooling.

We need alternatives of SRAM for designing cache

Die-stacked DRAM





"Vertical"

3D stacking

"Horizontal"

2.5D stacking

Loh et al. MICRO'11

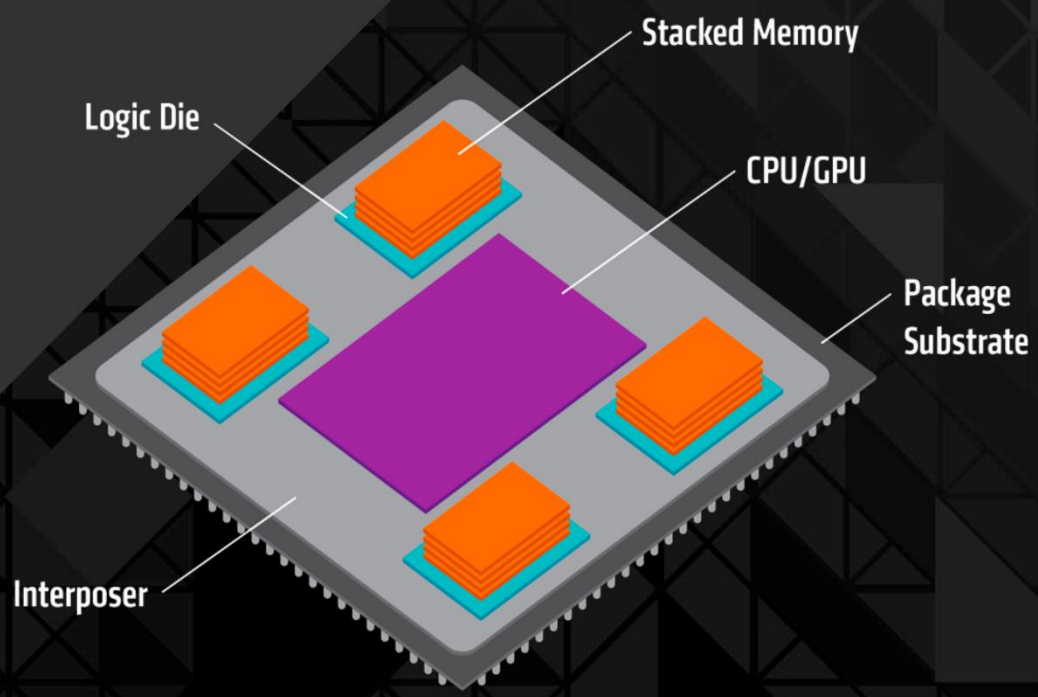
Example: HBM (high bandwidth memory)

THE INTERPOSER

THE NEXT STEP IN INTEGRATION

AMD

- Brings DRAM as close as possible to the logic die
- Improving proximity enables extremely wide bus widths
- Improving proximity simplifies communication and clocking
- Improving proximity greatly improves bandwidth per watt
- Allows for integration of disparate technologies such as DRAM
- AMD developed industry partnerships with ASE, Amkor & UMC to develop the first high-volume manufacturable interposer solution



Courtesy: AMD

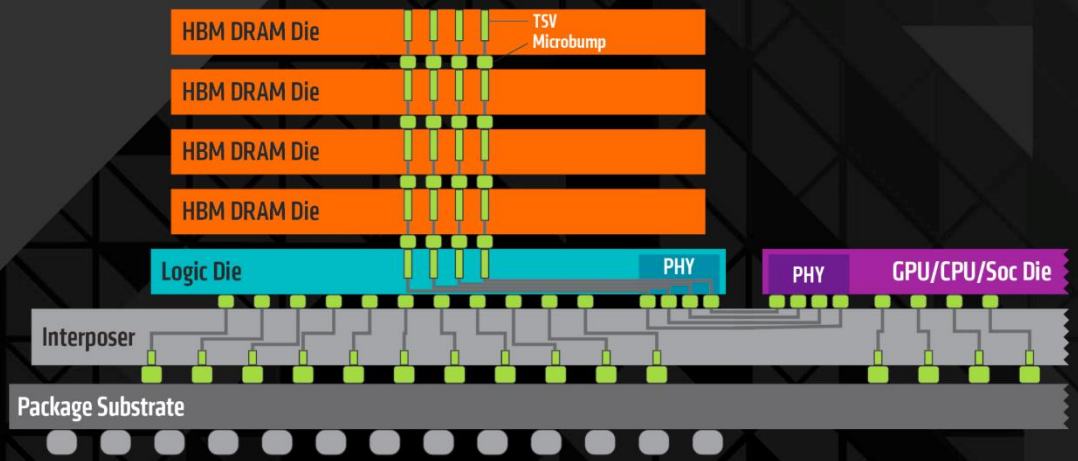
Example: HBM

HIGH-BANDWIDTH MEMORY

DRAM BUILT FOR AN INTERPOSER

AMD

- A new type of memory chip with low power consumption and an ultra-wide bus width
- Many of those chips stacked vertically like floors in a skyscraper
- New interconnects, called "through-silicon vias" (TSVs) and "µbumps", connect one DRAM chip to the next
- TSVs and µbumps also used to connect the SoC/GPU to the interposer
- AMD and SK Hynix partnered to define and develop the first complete specification and prototype for HBM



Sparsh Mittal, IIT Roorkee

Courtesy: AMD

Die-stacked DRAM technologies

- **Promising solution to the memory wall problem**
  - Interconnects logic and DRAM dies using TSVs (3D) and silicon interposers (2.5D)
  - Adopted by major processor and DRAM vendors for throughput and energy efficiency
    - Samsung, micron, SK Hynix, IBM, Intel, AMD, NVIDIA

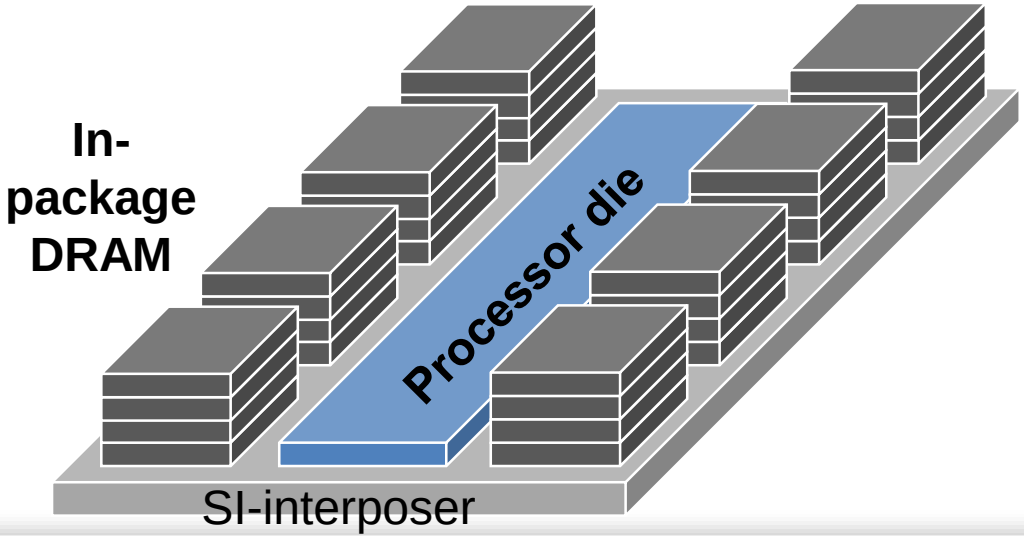


Figure credit: Lee et al. ISCA'15

Die-stacked DRAM technologies

- **Two options to use it**
  - As large DRAM cache Our focus
  - As part of main memory

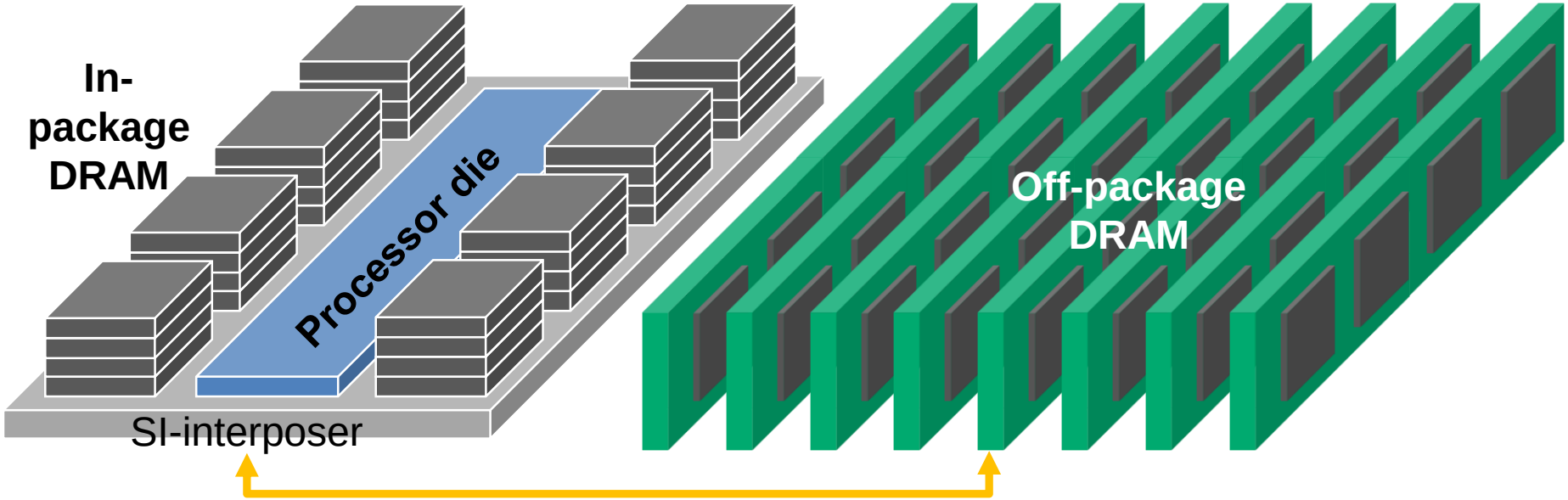
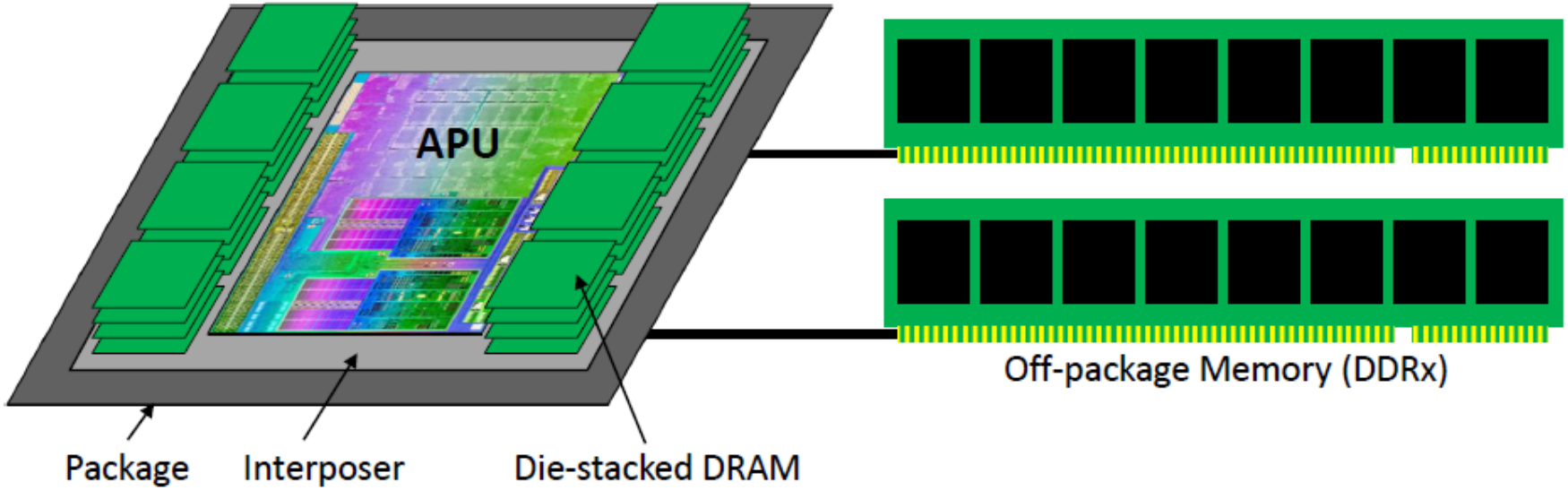


Figure credit: Lee et al. ISCA'15

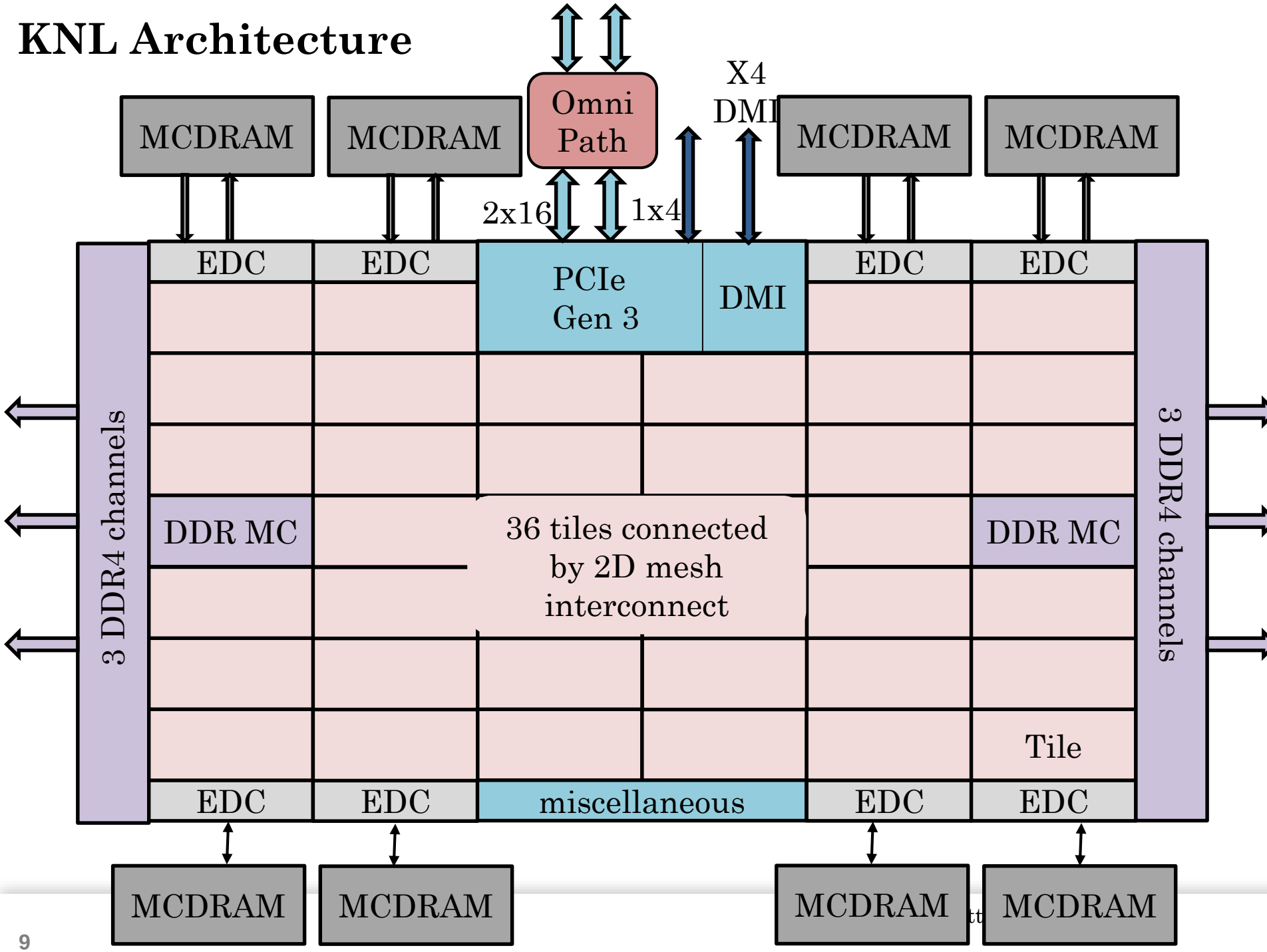
Overall system



2.5D-integrated DRAM stacks along with conventional off-package DDR memory

Figure credit: Meswani et al. HPCA'15

KNL Architecture



Benefits of Stacked DRAM

Benefits of Stacked DRAM

- Higher integration density
- 8-15X higher bandwidth than off-chip DRAM with only 1/2 to 1/3 of latency
- Reduces cache area, allows placing higher number of cores
- Integration in products: Intel's 14nm Knights Landing processor:
  - 16GB stacked DRAM last level cache with 500GB/s bandwidth and
  - 384GB off-chip DRAM with 90 GB/s bandwidth

Benefits of Stacked DRAM

- Energy per bit of TSV based interface= ~1/10X LPDDR2 and ~1/100X DDR3
- Stacking multiple processor chips may create thermal issues, but stacking memory on top of processor cores increases temperature only slightly (<10 degree C for 8 memory layers)
- On using stacked DRAM for cache, software need not be modified to reap benefits of cache

Challenges of Stacked DRAM

Capacity/latency tradeoff

- Provides smaller capacity than off-chip DRAM
- Stacked DRAM bandwidth is restricted by parallelism in stacked DRAM itself, not by interface.
- Hence, stacked DRAM is better fit as cache than as memory
- Poor locality app => DRAM cache may not reduce miss-rate
- DRAM cache latency: only 50% lower than main memory latency => they increase latency of misses
- Tradeoff between capacity and latency => cannot increase capacity beyond a point

Design Complexity

- LRU policy requires updating replacement info & reduces availability => have to use random replacement policy
- DRAM cache controller needs to perform device-level tasks e.g., managing refresh, issuing activation, precharge commands, fulfilling timing constraints, etc.
- => high complexity, requires redesign of cache control logic

Reliability Challenges

- Heat dissipation in difficult in 3D systems => thermal management issues
- 32ms refresh period may be required
- Faults in die-stacked memory may not be easily serviceable => may require discarding the entire package including the functioning processor layer
- Designing memory with very large number of layers => challenges of power delivery and cooling, yield, testability.

REFERENCES

- S. Mittal, “A Survey Of Techniques for Architecting DRAM Caches”, IEEE TPDS 2016 (pdf)